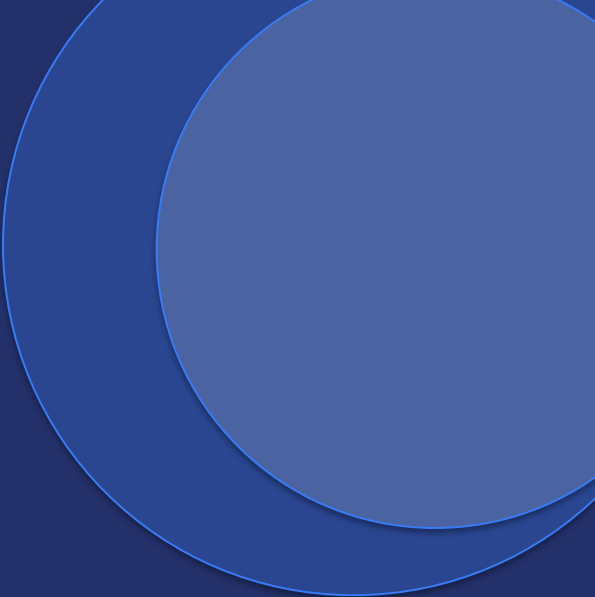




Banking Regulatory Intelligence:

GraphRAG Phase 1

OpenAI-Based Proof of Concept for Graph-Enhanced Retrieval, Evidence Grounding, Runtime Guardrails, and Public Deployment

Microsoft GraphRAG 3.1.1 | GPT-4.1 | 23 Banking & Regulatory Documents

Jingru Chen • Phase 1 Complete — August 31, 2026

Executive Summary



THE PROBLEM

Banking regulatory and model-risk questions often span multiple documents, institutions, entities, and relationships. Traditional retrieval can return relevant passages but may struggle with corpus-wide synthesis and evidence completeness.



THE SOLUTION

A Microsoft GraphRAG Phase 1 PoC using GPT-4.1, graph/community indexing, Basic + Local + Global Search, evidence-grounded citations, runtime guardrails, and a Streamlit public application.



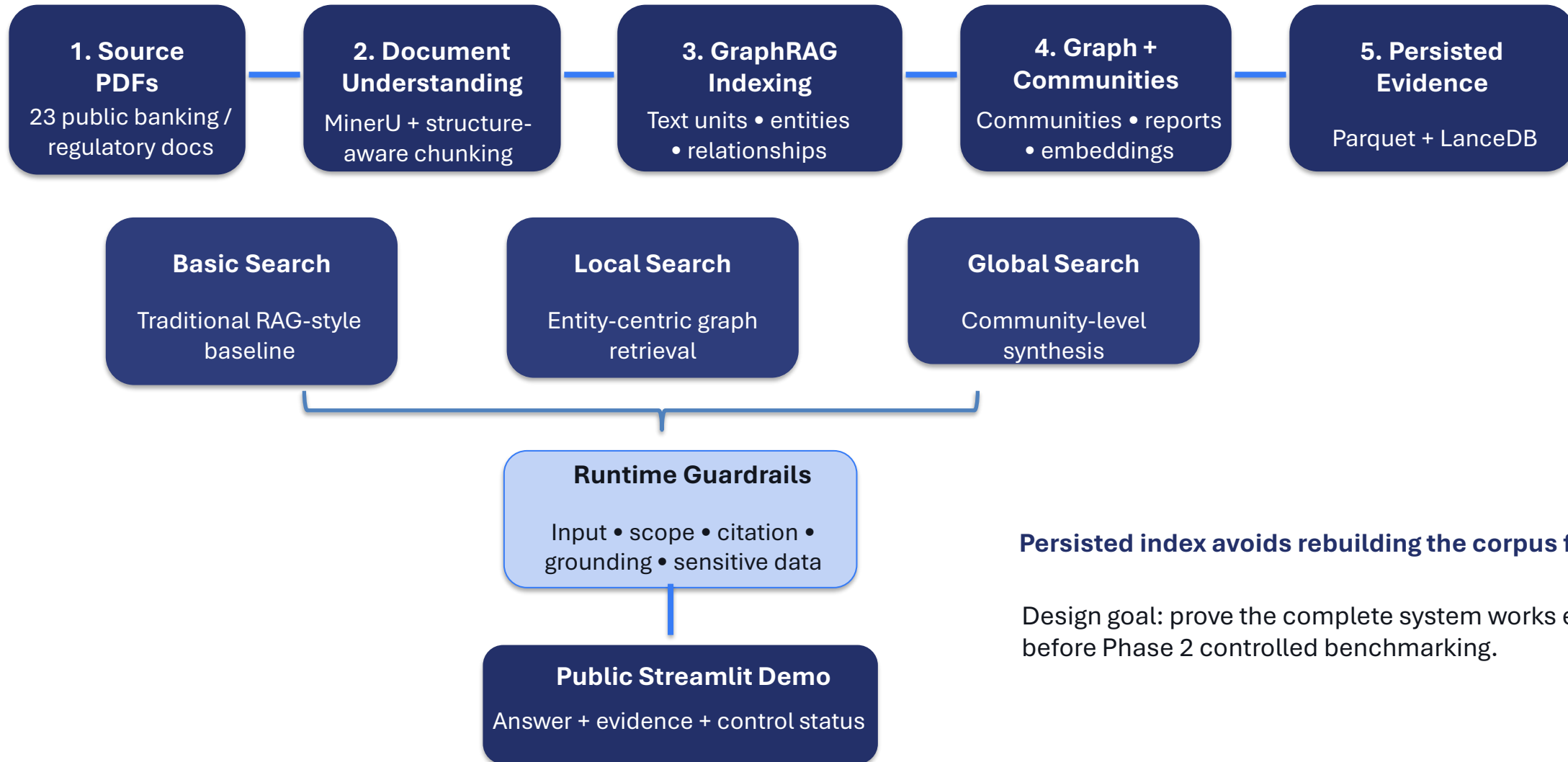
THE IMPACT

- **End-to-end GraphRAG system proven**
- Public portfolio deployment
- Guardrail PASS/WARN/BLOCK behavior
- Global Search reduced from ~52 min to ~8.8 min public
- Known retrieval gaps converted into Phase 2 requirements

Public Demo: <https://banking-graphrag-jingru.streamlit.app>

GitHub: <https://github.com/jxc518/graphrag-banking-regulatory-intelligence/blob/main/README.md>

GraphRAG Phase 1 Architecture



Persisted index avoids rebuilding the corpus for every query.

Design goal: prove the complete system works end-to-end before Phase 2 controlled benchmarking.

Document Intelligence & GraphRAG Indexing Pipeline

1

PDF → Markdown

MinerU normalizes layout-heavy regulatory PDFs and bank quarterly earnings supplement files while preserving source structure.

2

Structure-Aware Chunks

Chunk boundaries preserve headings, narrative blocks, tables, and evidence granularity.

3

Entity / Relationship Extraction

LLM extracts graph concepts and relationships from text units.

4

Community Detection

Graph communities organize related entities and relationships hierarchically.

5

Community Reports + Embeddings

Summaries and vector representations support Local / Global retrieval.

Observed Full-Index Workload

- 23 source PDFs processed
- 1,743 / 1,743 graph extraction units completed
- 22,963 entity / relationship description summarization tasks observed
- 18.3M tokens and 8,000+ API requests consumed
- Hierarchical community summarization across multiple levels
- Index artifacts persisted to Parquet and LanceDB for query-time reuse

Basic vs Local vs Global Search

BASIC SEARCH

Traditional RAG-style baseline

Best for:

- direct document questions
- quick baseline comparison
- passage-level retrieval

Trade-off: limited graph reasoning across entities.

LOCAL SEARCH

Entity-centric graph retrieval

Best for:

- institution / entity questions
- relationships and neighborhood context
- graph-enhanced evidence

Trade-off: multi-entity coverage can be uneven.

GLOBAL SEARCH

Community-based corpus synthesis

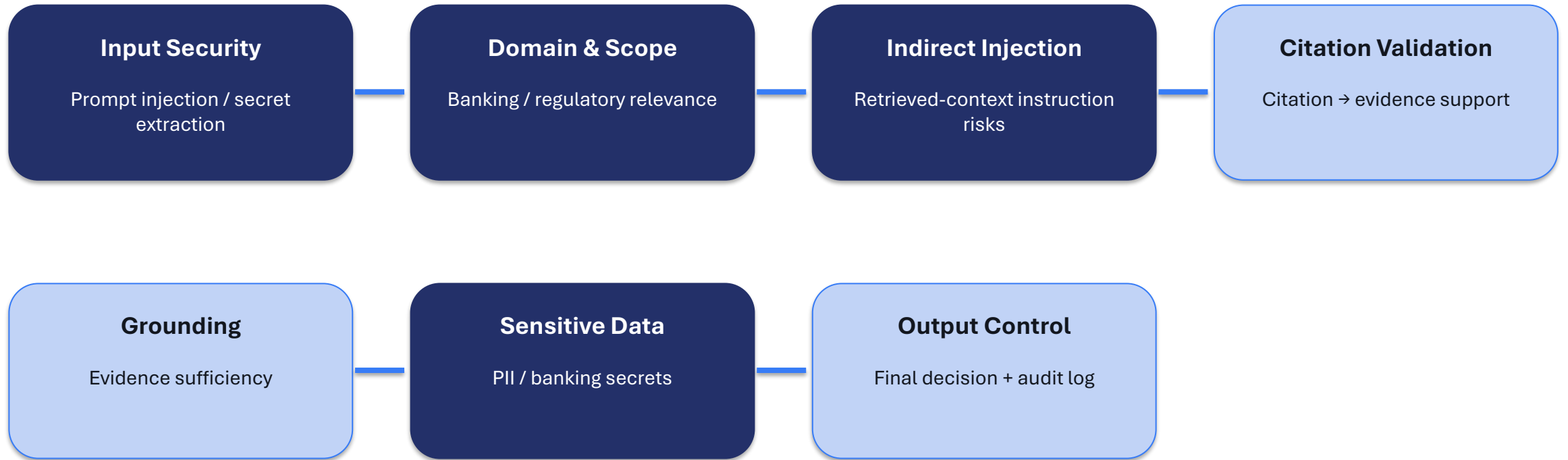
Best for:

- cross-document themes
- broad comparative questions
- corpus-wide synthesis

Trade-off: highest latency and cost.

Key design principle: select retrieval mode by question class — not one mode for every query.

Runtime Guardrail Architecture



PASS

WARN

BLOCK

Governance principle: WARN is preserved when evidence support is incomplete; thresholds are not lowered merely to make results green.

Final Public Benchmark Evidence

Search Mode	Runtime	Evidence Refs	Guardrail
Basic	23.9 sec	21	PASS
Local	25.6 sec	55	PASS
Global	527.7 sec (~8.8 min)	180	WARN

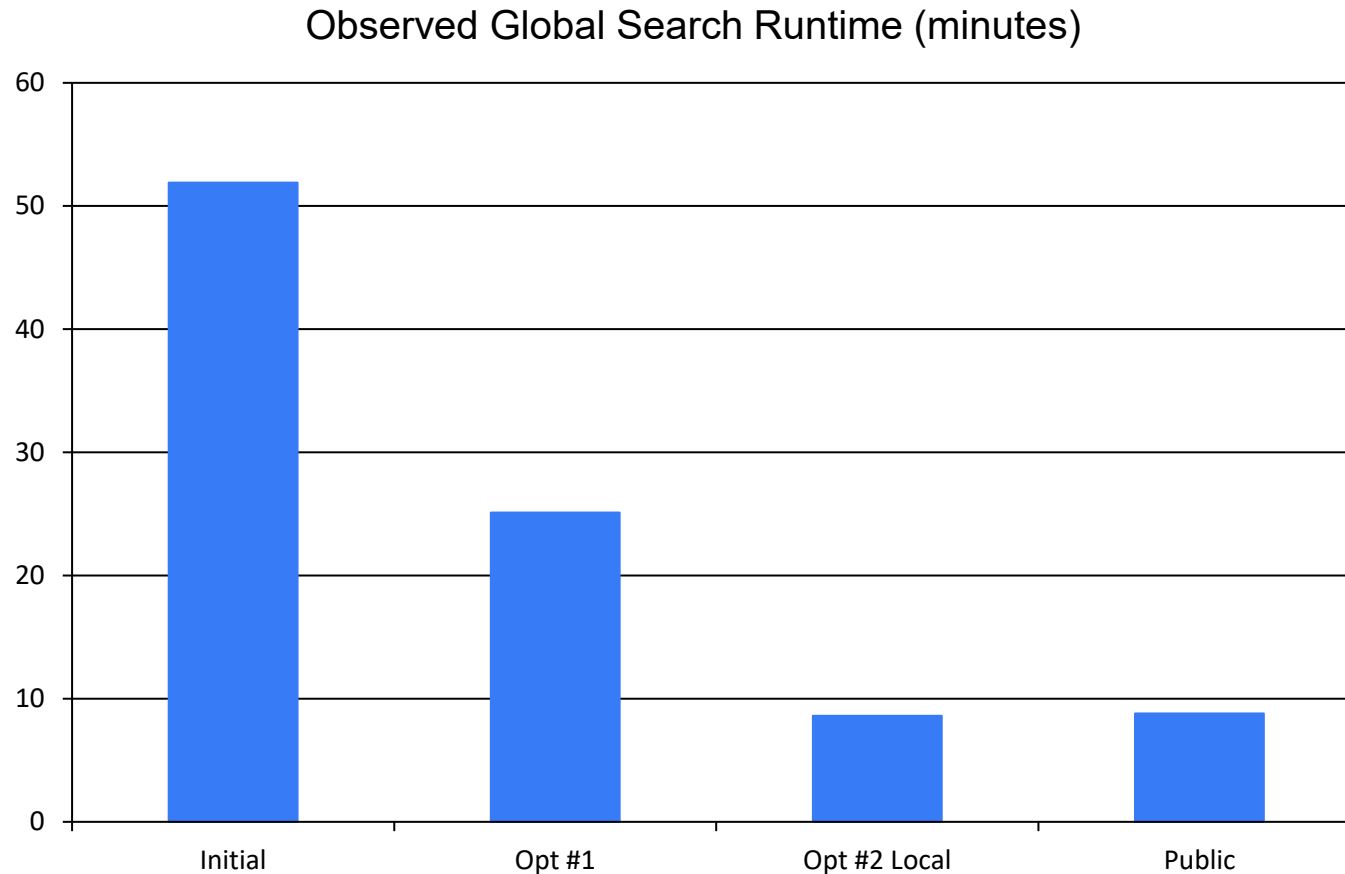
WHAT THE BENCHMARK PROVES

All three public query modes execute end-to-end with visible evidence and runtime governance. The public Global run remained under 9 minutes after optimization.

WHAT IT DOES NOT PROVE

Phase 1 is qualitative / demo-oriented. It does not claim statistical superiority, deterministic quality equivalence, or optimal retrieval architecture.

Global Search Performance Optimization



≈83% latency reduction vs. the original successful Global Search run

Optimization Path

- Community level: 2 → 1
- Dynamic community selection: OFF → ON
- OpenAI throughput: 24K TPM / 8 RPM
→ 60K TPM / 20 RPM
- GPT-4.1 and indexed corpus held constant
- Final public runtime: ~8.8 minutes

Engineering Findings & Debugging Lessons

1. Retrieval ≠ Deterministic

Same query can produce different evidence counts and claim sets; evaluation must tolerate controlled variability.

2. Citations ≠ Support

A citation can exist while a claim is only partially supported; citation-to-evidence validation matters.

3. Numeric False Positives

Enumeration markers such as (1) and (2) can look like numeric claims; enumeration-aware handling fixed the observed regression.

4. Identifier Discoverability

Exact identifiers such as SR 11-7 exposed alias / canonical-term retrieval gaps.

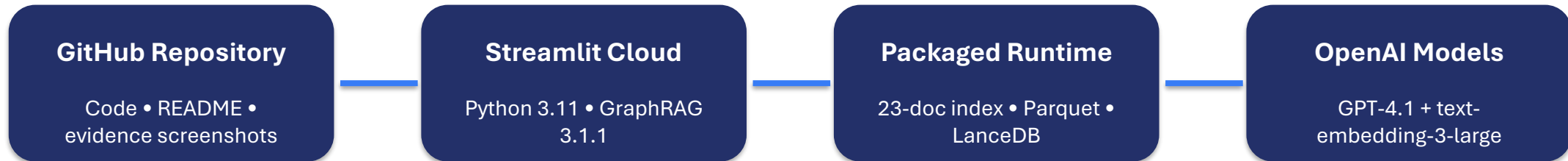
5. Multi-Entity Completeness

Cross-bank comparisons can retrieve uneven evidence; entity-specific retrieval and completeness gating are needed.

6. Long-Running Operations

Global Search requires production patterns such as caching, async execution, job state, and resilience.

Public Deployment & Governance Controls



Security & Secret Handling

- No hard-coded API key in tracked source
- GRAPH_RAG_API_KEY injected through Streamlit Secrets / environment variable
- ``.env`` and `secrets.toml`` excluded from Git
- Final GitHub security / secret audit passed

Public Cost & Abuse Controls

- Maximum query length: 500 characters
- Basic / Local: max 10 successful runs per session
- Global: max 2 successful runs per session
- Global cooldown: 10 minutes
- Backend 20 RPM / 60K TPM is separate from public-app limits

Public demo: banking-graphrag-jingru.streamlit.app

Phase 2 Roadmap — From PoC to Production-Grade GraphRAG

Capability Area	Phase 1 Finding	Phase 2 Enhancement
Agentic GraphRAG & Routing	Users manually choose Basic / Local / Global.	Add planning, routing, validation, synthesis, and governance agents.
Multi-Entity Retrieval & Resolution	Uneven evidence coverage and exact-identifier gaps.	Use query decomposition, entity-specific retrieval, and canonical alias resolution.
OpenAI vs DeepSeek Comparison	Provider differences appeared in structured output, stability, latency, and errors.	Benchmark quality, latency, cost, errors, structured-output success, and retry recovery.
Stronger Guardrails & Evidence	PASS/WARN/BLOCK works, but claim-level support remains challenging.	Strengthen injection defense, citation/numeric validation, PII controls, and abstention.
Provider & Model Abstraction	Provider behavior is not fully interchangeable.	Add structured-output adapters, provider policies, and task-level model routing.
Failure Recovery & Reliability	Individual inference tasks can fail while the pipeline continues.	Add failed-task registry, targeted retries, recovery, and error classification.
Quantitative Evaluation	Phase 1 evaluation is mainly qualitative and demo-oriented.	Benchmark retrieval, grounding, citations, guardrails, latency, cost, and stability.
Performance & Cost Governance	Global Search remains compute- and latency-intensive.	Add caching, precomputation, async execution, and budget-aware routing.
Production Observability & Security	PoC audit logs and session controls are limited.	Add P95 metrics, failure/cost telemetry, distributed rate limits, identity, and security controls.

Phase 2: reliable, secure, measurable, and cost-effective GraphRAG across models and providers.

Appendix 1: Evidence #1 — MinerU Regulatory Document Understanding

MinerU transforms layout-heavy regulatory PDFs into structured, machine-readable Markdown, preserving document hierarchy for downstream structure-aware chunking and GraphRAG indexing.

3. Structured Markdown

• PS C:\Users\chen_\Documents_67_2026_Job_Hunting_After (& c:\Users\chen_\graphrag\.venv\Scripts\Activate.ps: (graphrag-monorepo) PS C:\Users\chen_\Documents_67_2 ob_Hunting_After_Wells_Fargo_67_39_RAG_GraphRAG_Agen April 4, 2011

SUPERVISORY GUIDANCE ON MODEL RISK MANAGEMENT

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I. Introduction
II. Purpose and Scope .2
III. Overview of Model Risk Management. .3
IV. Model Development, Implementation, and Use. .5
V. Model Validation . .9
VI. Governance, Policies, and Controls. .16
VII. Conclusion. .21

I. INTRODUCTION

Banks rely heavily on quantitative analysis and models in most aspects of financial decision making.¹ They routinely use models for a broad range of activities, including underwriting credits; valuing exposures, instruments, and positions; measuring risk; managing and safeguarding client assets; determining capital and reserve adequacy; and many other activities. In recent years, banks have applied models to more complex products and with more ambitious scope, such as enterprise-wide risk measurement, while the markets in which they are used have also broadened and changed. Changes in regulation have spurred some of the recent developments, particularly the U.S. regulatory capital rules for market, credit, and operational risk based on the framework developed by the Basel Committee on Banking Supervision. Even apart from these regulatory considerations, however, banks have been increasing the use of data-driven, quantitative decision-making tools for a number of years.

1. Original Regulatory PDF

SR Letter 11-7
Attachment

Board of Governors of the Federal Reserve System
Office of the Comptroller of the Currency

April 4, 2011

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2. Layout-Aware Parsing

SR Letter 11-7
Attachment

Board of Governors of the Federal Reserve System
Office of the Comptroller of the Currency

April 4, 2011¹

SUPERVISORY GUIDANCE ON MODEL RISK MANAGEMENT²

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Appendix 2: Evidence #2 — Knowledge Graph Persistence & Relationship Semantics

← ↻ ⓘ localhost:7474/browser/ ☆ ☆ ⋮ ⋮ Chat

● Instance: neo4j://localhost:7687 ▾ Database: neo4j CYPHER 25 ▾ User: neo4j ▾ [Connect your instance to Aura console for more tools and features ↗](#)

Database information

Nodes (210)

* __Chunk__ __Community__

__Document__ __Entity__ __Finding__

__Relationship__ Event Organization Person

Relationships (311)

* HAS_ENTITY HAS_FINDING

HAS_RELATIONSHIP IN_COMMUNITY

PART_OF RELATED

Property keys

age children city combined_degree

community creation_date degree

description explanation frequency

full_content human_readable_id id

import_timestamp industry level location

n_tokens name parent period rank

rating_explanation relationship_id size

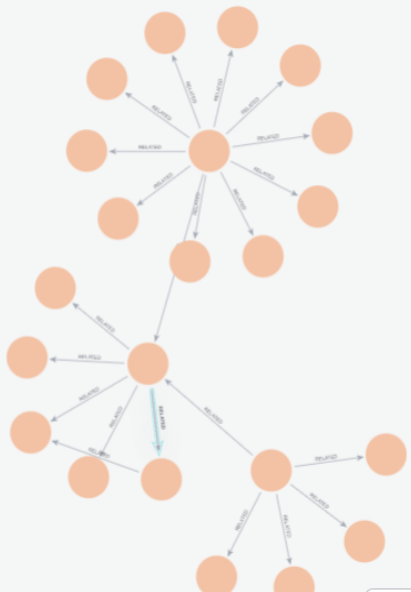
summary text title weight

^ Show less

neo4j\$

neo4j\$ MATCH (a: `\_\_Entity\_\_`)-[r:RELATED]-(b: `\_\_Entity\_\_`) RETURN a, r, b LIMIT 25

Graph Table Raw



Relationship details

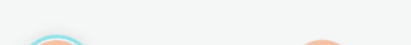
RELATED

Key	Value
<id>	5:fc975d23-7b11-4718-97bb-467bfd3ab7a8:57
description	"Credit risk models integrate probability of default, loss given default, and exposure at default to estimate expected credit loss"
relationship_id	"79fe0dda-472b-40a5-9036-5b695ed53381"
weight	10.0

Started streaming 25 records after 41 ms and completed after 43 ms.

neo4j\$ MATCH (e: `\_\_Entity\_\_`) RETURN e LIMIT 5

Graph Table Raw



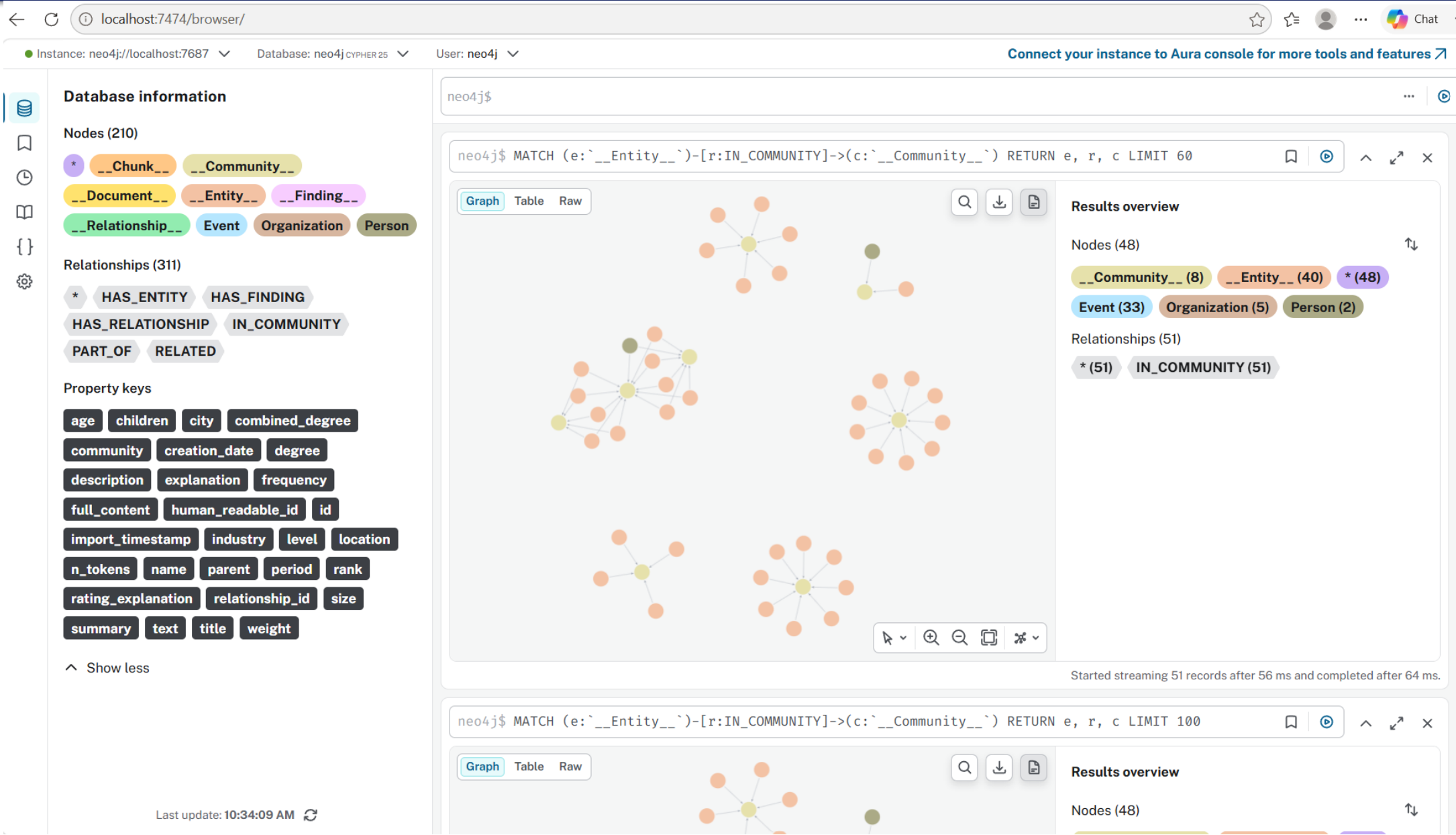
Node details

__Entity__ Event

Last update: 10:34:09 AM ↻

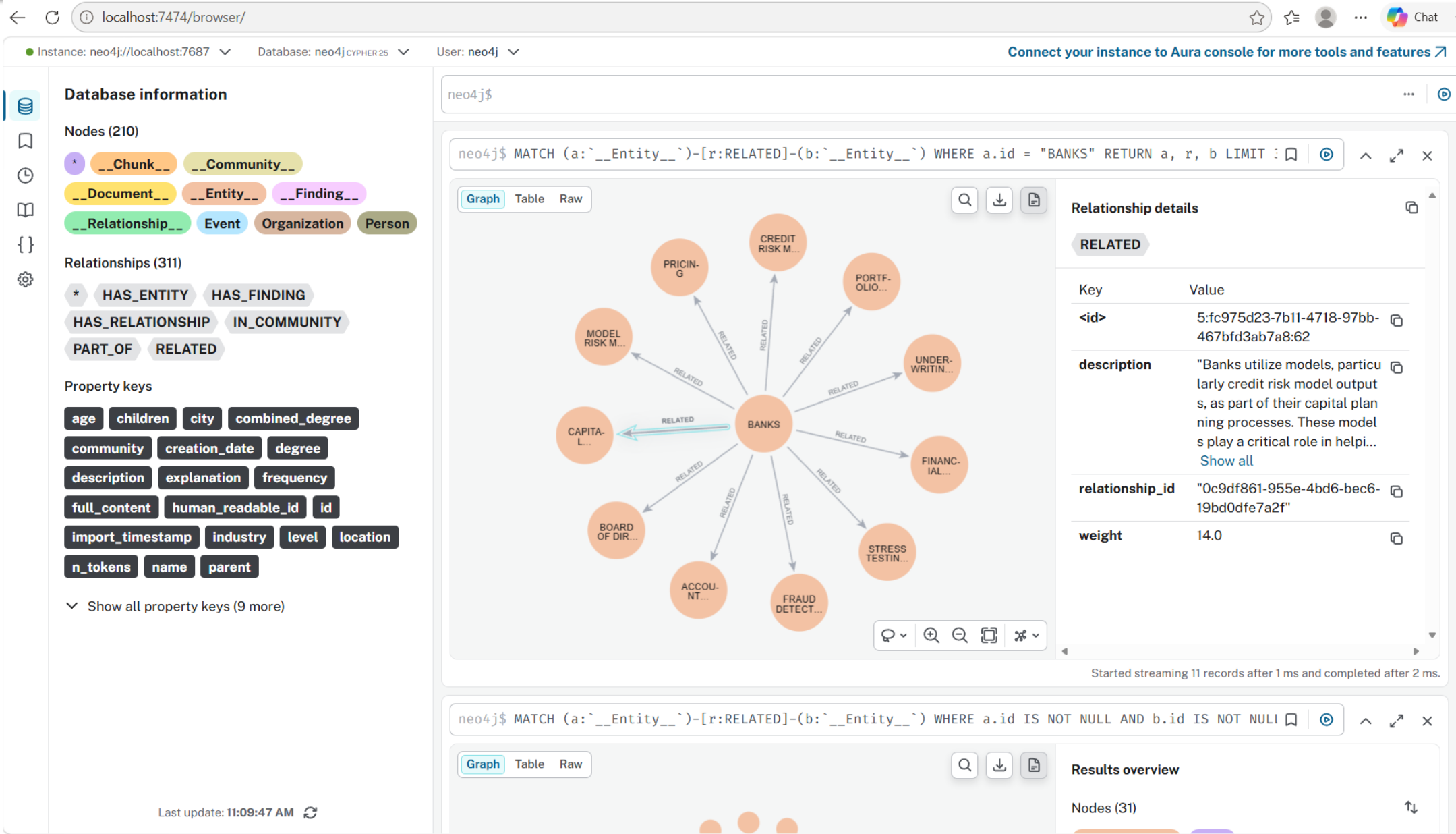
GraphRAG-extracted entities and weighted semantic relationships persisted in Neo4j, with relationship-level descriptions supporting traceable graph inspection.

Appendix 3: Evidence #3 — Hierarchical Community Structure



GraphRAG organizes extracted entities into higher-level communities, enabling corpus-wide thematic synthesis for Global Search.

Appendix 4: Evidence #4 — BANKS-Centered Banking Relationship Subgraph



Thank You

Questions & Discussion
